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Recap

Admittance parameters

Short circuit admittance

$$y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}$$

$$y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$

transfer parameters

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$



$$[Z]_{N\times N} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow Y = \left[[Z]^{-1} \right]_{2\times 2}$$

1st 2 rows & columns

last class: simple examples → R/C/L

all these case $y_{12} = y_{21}$ (reciprocity thm)

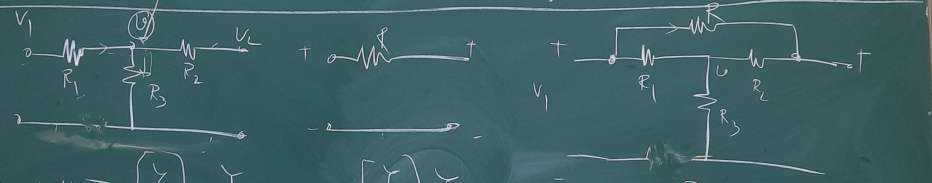
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3rd proof of Reciprocity theorem

$([Z]_{ij} = [Z]_{ji}$ if all passive)

$$\Rightarrow \left[Z^{-1}\right]_{ij} = \left[Z^{-1}\right]_{ji}$$

use KCL to get it



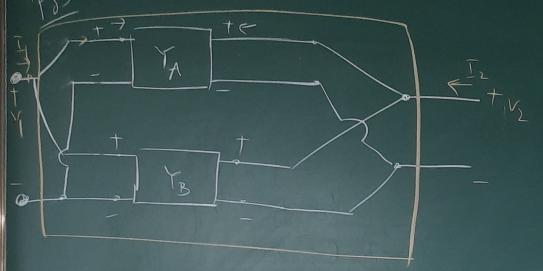
$$I_1 = y_{11}V_1 + y_{12}V_2$$

$$I_2 = y_{21}V_1 + y_{22}V_2$$

$[Y]$ Y_A $[Y]$ Y_B ? $Y_A + Y_B$

Remark: Conductance/admittance add up ∥

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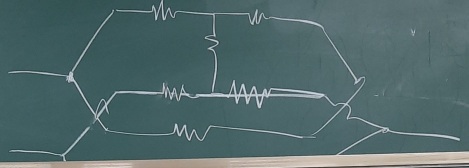


$Y = Y_A + Y_B$

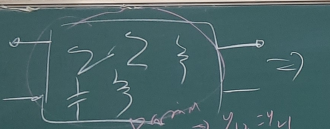


currents add up

Voltage diff same

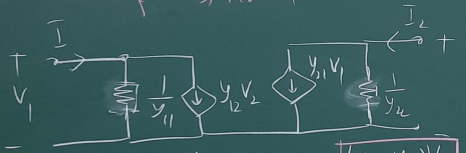


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$\Rightarrow$ [Y]

passive $\Rightarrow y_{12} = y_{21}$

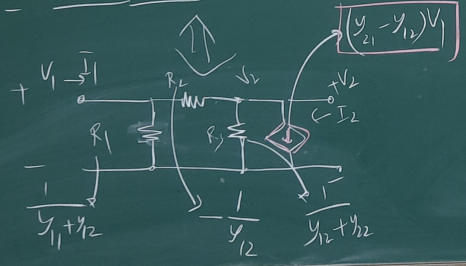


$$I_1 = (y_{11} + y_{12})V_1 - y_{12}(V_1 - V_2)$$

$$= (y_{11} + y_{12})V_1 - y_{12}V_1 + y_{12}V_2$$

$$y_{11}V_1 + y_{12}V_2$$

Any large passive Circuit $\Rightarrow$ reduced to smaller pass circ



$$I_1 = \frac{V_1}{R_1} + \frac{V_1 - V_2}{R_2}$$

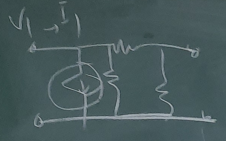
$$I_2 = \frac{V_2}{R_3} + \frac{V_2 - V_1}{-\frac{1}{y_{12}}} + (y_{21} - y_{12})V_1$$

$$I_2 = V_2 y_{22} + V_1 y_{21}$$

Chapter 16

$\frac{1}{R_3} = Y_{12} = Y_{21}$

$\Rightarrow \frac{1}{R_3} = Y_{12} + Y_{22}$



$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

P2C

$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = [Y] \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \Rightarrow \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = [Y]^{-1} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$

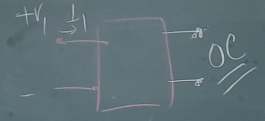
$[Z]$ impedance matrix

$[Z]$ NXN $\neq$ $[Z]$

Find 2 rows & columns

$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2 = 0}$ OC i/p impedance

$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1 = 0}$



Z_{22} = OC o/p impedance

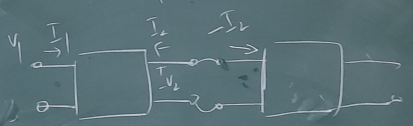
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$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

$I_1 = Y_{11} V_1 + Y_{12} V_2$

$I_2 = Y_{21} V_1 + Y_{22} V_2$



$\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$ in terms of $\begin{matrix} V_1 \\ I_1 \end{matrix}$

$\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$

$\Updownarrow$

$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$

Devices modelling

$\begin{matrix} V_1 \\ I_2 \end{matrix}$ in terms of $\begin{matrix} V_2 \\ I_1 \end{matrix}$

$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = [h] \begin{bmatrix} V_2 \\ I_1 \end{bmatrix}$

Hybrid